

Vineet Gawali

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer and Final year AI & DS student who designs and ships agentic systems, not just prototypes — most recently Local GraphRAG, a fully local multi-agent RAG pipeline combining vector and knowledge-graph retrieval with zero cloud dependency. NVIDIA-certified in RAG agent development, with hands-on Python, PyTorch, and FastAPI experience. Seeking a Generative AI Engineering internship focused on agentic systems and retrieval-augmented architectures.

TECHNICAL SKILLS

- **Generative AI & Agentic Systems:** LangGraph (multi-agent orchestration), RAG Pipelines (vector + knowledge-graph), LLM APIs (Groq, OpenAI, local via LM Studio), Prompt Engineering, LLM Application Security (OWASP LLM Top 10, jailbreak defense), Vector Databases (ChromaDB, FAISS).
- **Backend & Deployment:** FastAPI, Streamlit, Flask, Docker, Git, CI/CD, AWS / Azure / GCP (basics).
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Diffusion Models, Computer Vision (OpenCV).
- **Programming Languages:** Python, Java, JavaScript.
- **Databases:** SQL(MySQL), MongoDB

CERTIFICATIONS

- [Model Context Protocol: Advanced Topics](#) – Anthropic (Issued Aug 2026)
- [Building RAG Agents with LLMs](#) – NVIDIA (Issued Jan 2026)
- **Generative AI with Diffusion Models** – AWS Training & Certification (Issued Sep 2025)
- [Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate](#) – Oracle University (Issued Sep 2025)
- [Generative AI with Diffusion Models](#) – NVIDIA (Issued Sep 2025)

PROJECTS

Local GraphRAG — Python, LangGraph, FastAPI, Streamlit, ChromaDB, LM Studio

- Architected a fully local, privacy-preserving multi-agent RAG system combining vector similarity search (ChromaDB) with knowledge-graph traversal, eliminating cloud LLM dependency by running 100% of inference on-device.
- Designed a 4-node LangGraph state machine that classifies each query with an LLM router and dynamically directs it to top-5 vector retrieval or 2-hop knowledge-graph traversal, then synthesizes grounded answers with a full reasoning trace.
- Built an automated knowledge-graph construction pipeline using LLM-based triple extraction, with content-hash deduplication and incremental merging to keep the graph consistent across repeated document uploads.
- Shipped a 7-module, ~2,150-line full-stack application (FastAPI backend with 4 REST endpoints, Streamlit frontend with real-time ingestion progress) delivering 18 validated end-to-end features, optimized to run within 4GB VRAM on a quantized Mistral-7B model.

AI Portfolio Assistant — LLM Security Hardening | Cloudflare Workers, Groq API (Llama 3.3 70B), JavaScript

- Red-teamed a self-built portfolio chatbot against 9 jailbreak/prompt-injection techniques mapped to the OWASP Top 10 for LLM Applications, uncovering 2 exploitable vulnerabilities (instruction override, system-prompt leakage) on the first test pass.
- Closed the override exploit by rewriting the system prompt from a denylist to an allowlist-scoped instruction hierarchy, and fixed the leakage vulnerability with a non-negotiable no-disclosure rule plus a deterministic output-side filter as defense-in-depth.
- Hardened the Cloudflare Worker middleware with an explicit CORS origin allowlist, per-IP rate limiting via Cloudflare KV, and a minimized response schema — closing quota-drain and schema-leakage risks and reaching 7/7 blocked techniques on the final pass.

AI Interviewer — React, TypeScript, Vite, Groq API, LLM Integration

- Designed a multi-step LLM reasoning loop that generates technical questions, evaluates candidate responses, and adapts follow-up questions based on performance — an early agentic pattern applied to interview simulation.
- Built and ran a blind LLM-as-Judge evaluation harness (G-Eval methodology, chain-of-thought rubric scoring) to validate prompt improvements on the Data Science interview track — pass rate rose from 92% to 100% across 155 questions.

EDUCATION

Bachelor of Engineering in **Artificial Intelligence and Data Science**

Government Engineering College Nargund (VTU) | 2023 – 2027